Chunk influence on the RAG predictions

Labels: **final**. Run: `/Users/janneslampe/Desktop/Coding/Master Thesis/Evaluation/RAG Analysis/LLM judge results/families`. Reviewed tier: llama_70b, mistral_14b, qwen_3b.

Party positions (median ground truth of the party's texts): DIE LINKE 1.9, BÜNDNIS 90/DIE GRÜNEN 2.8, SPD 3.1, FDP 4.6, CDU/CSU 4.9, AfD 6.6

Chunks without a position (excluded from pull / distance): Fraktionslos 23, BSW 14, UNKNOWN 3

Shares of human-checked R_top labels (review, spot check, consensus, transfer): llama_3b 26%, llama_70b 74%, llama_8b 38%, mistral_14b 78%, mistral_3b 24%, mistral_8b 26%, qwen_32b 17%, qwen_3b 79%, qwen_72b 25%, qwen_7b 28%

Tier: reviewed (3 models, 268 texts, 1340 chunks)

Texts per quadrant: Q1 55, Q2 73, Q3 75, Q4 65

1. Quadrant profiles (means over texts)

feature	Q1	Q2	Q3	Q4	p_kruskal	p_Q1_vs_Q2
mean R_top	2.60	2.92	2.83	2.99	0.388	0.189
share chunks R_top ≥ 4	0.24	0.34	0.34	0.36	0.566	0.297
mean R_ideo	3.85	3.87	3.79	3.82	0.663	0.608
share chunks R_ideo ≥ 4	0.74	0.75	0.67	0.70	0.362	0.489
chunks cited (of 5)	3.04	2.55	3.07	3.03	0.315	0.180
R_top of cited chunks	2.80	3.08	2.90	3.14	0.440	0.344
R_ideo of cited chunks	3.90	3.84	3.78	3.89	0.526	0.847
share chunks from input's party	0.29	0.30	0.28	0.33	0.871	0.864
share cited chunks from input's party	0.40	0.36	0.26	0.42	0.064	0.253
	cited position – ground truth		0.99	0.93	1.44	0.84

2. Utilisation: share of chunks cited (A_caus = 1) by label level

dimension	level	n	cited_all	cited_Q1	cited_Q2	cited_Q3	cited_Q4
R_top	1	207	0.44	0.58	0.27	0.63	0.24
R_top	2	472	0.57	0.50	0.51	0.61	0.62
R_top	3	227	0.62	0.65	0.59	0.57	0.67

dimension	level	n	cited_all	cited_Q1	cited_Q2	cited_Q3	cited_Q4
R_top	4	189	0.67	0.79	0.57	0.69	0.64
R_top	5	245	0.63	0.70	0.58	0.56	0.71
R_ideo	2	14	0.43	—	0.50	0.60	0.20
R_ideo	3	367	0.58	0.61	0.54	0.64	0.51
R_ideo	4	789	0.59	0.59	0.50	0.63	0.63
R_ideo	5	170	0.59	0.68	0.48	0.49	0.75

3. Pull: does the RAG prediction move toward the evidence?

pull = mean party position of the chunks − baseline prediction; shift = RAG − baseline prediction. ρ > 0 and slope > 0 mean the prediction follows the evidence; slope 1 = it moves all the way.

subset	n_texts	spearman_shift_vs_pull	p	slope	share_moved_toward_evidence
all texts	233	0.54	<0.001	0.70	0.47
Q1 (RAG win)	50	0.86	<0.001	1.17	0.84
Q2 (RAG distraction)	58	0.66	<0.001	1.10	0.64
Q3 (joint failure)	68	0.30	0.013	0.19	0.37
Q4 (baseline sufficient)	57	-0.26	0.054	-0.04	0.07
cited chunks mostly relevant (R_top ≥ 4)	71	0.47	<0.001	0.60	0.41
cited chunks less relevant (R_top < 4)	162	0.56	<0.001	0.73	0.49
cited chunks ideological (R_ideo ≥ 4)	134	0.53	<0.001	0.75	0.42

subset	n_texts	spearman_shift_vs_pull	p	slope	share_moved_toward_evidence
cited chunks less ideological (R_ideo < 4)	99	0.49	<0.001	0.59	0.53

Same with all retrieved chunks (cited or not):

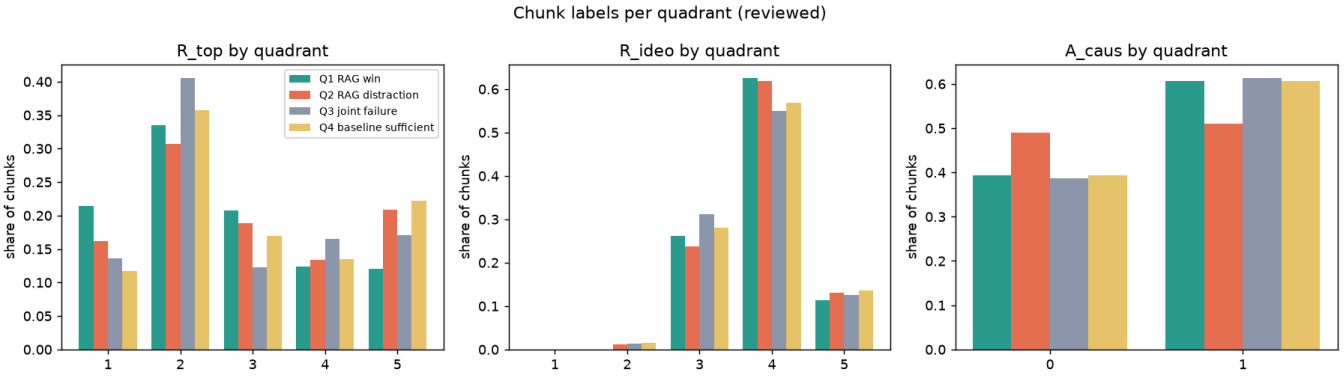
subset	n_texts	spearman_shift_vs_pull	p	slope	share_moved_toward_evidence
all texts	268	0.56	<0.001	0.73	0.50

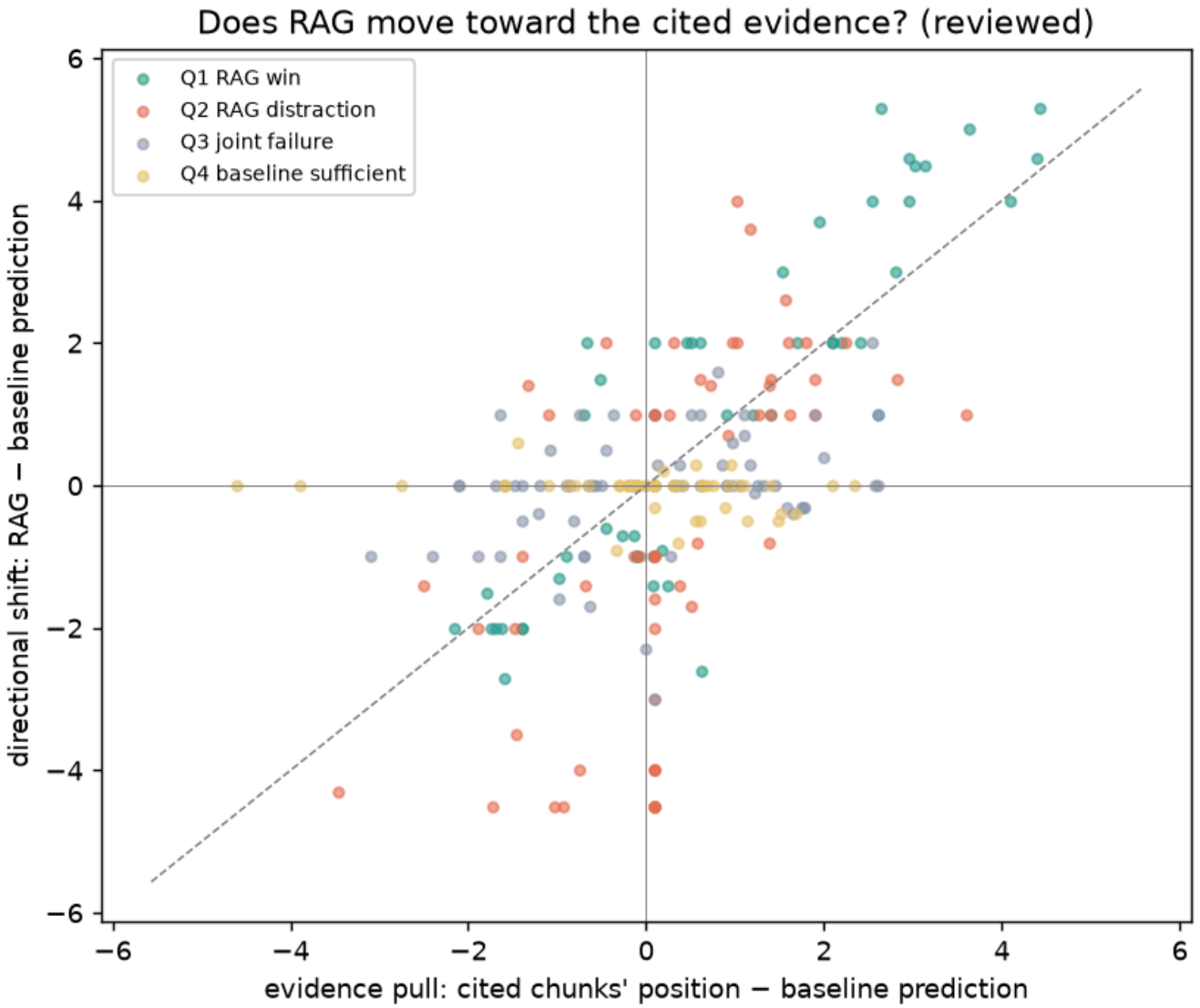
4. Evidence vs. outcome (Spearman with error reduction = base error – RAG error)

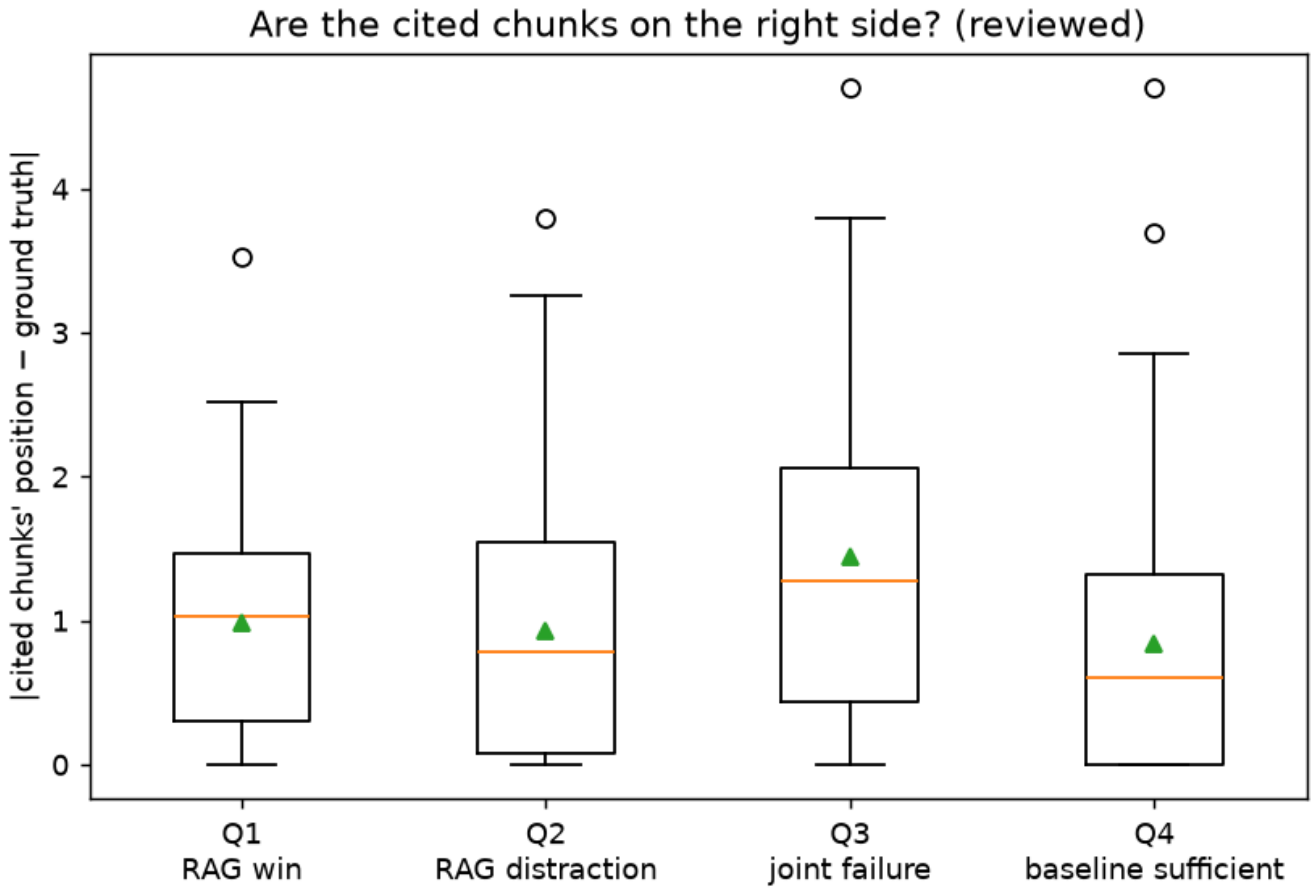
feature	n_texts	spearman_vs_error_reduction	p
cited chunks' distance to ground truth	233	0.01	0.911
share cited chunks from input's party	234	0.08	0.253
R_top of cited chunks	234	-0.07	0.275
R_ideo of cited chunks	234	-0.02	0.780
mean R_top (all chunks)	268	-0.08	0.170
mean R_ideo (all chunks)	268	-0.06	0.338
chunks cited	268	0.06	0.319
share chunks from input's party	268	0.02	0.704

Per model (Q1 vs. Q2)

model	tier	texts	Q1 cited R_top	Q1 cited–GT	Q2 cited R_top	Q2 cited–GT	ρ(shift, pull)
llama_70b	reviewed	97	2.88	0.56	3.19	0.63	0.62
mistral_14b	reviewed	100	2.67	1.28	3.15	1.09	0.57
qwen_3b	reviewed	71	3.10	1.26	2.56	1.25	0.49


Chunk labels per quadrant (reviewed)
R_top by quadrant
R_ideo by quadrant
A_caus by quadrant
share of chunks
Q1 RAG win
Q2 RAG distraction
Q3 joint failure
Q4 baseline sufficient


Does RAG move toward the cited evidence? (reviewed)
Q1 RAG win
Q2 RAG distraction
Q3 joint failure
Q4 baseline sufficient
directional shift: RAG − baseline prediction
evidence pull: cited chunks' position − baseline prediction



Tier: all (10 models, 968 texts, 4840 chunks)

Texts per quadrant: Q1 230, Q2 248, Q3 250, Q4 240

1. Quadrant profiles (means over texts)

feature	Q1	Q2	Q3	Q4	p_kruskal	p_Q1_vs_Q2
mean R_top	2.80	2.99	2.92	2.90	0.321	0.064
share chunks R_top ≥ 4	0.33	0.38	0.38	0.36	0.604	0.222
mean R_ideo	3.77	3.81	3.78	3.81	0.785	0.436
share chunks R_ideo ≥ 4	0.68	0.71	0.68	0.71	0.481	0.394
chunks cited (of 5)	2.85	2.60	2.89	2.95	0.162	0.200
R_top of cited chunks	2.92	3.13	3.07	3.01	0.342	0.089
R_ideo of cited chunks	3.80	3.81	3.81	3.85	0.591	0.549
share chunks from input's party	0.33	0.26	0.24	0.36	<0.001	0.003
share cited chunks from input's party	0.43	0.30	0.27	0.46	<0.001	<0.001
	cited position – ground truth		0.84	1.13	1.31	0.90

2. Utilisation: share of chunks cited (A_caus = 1) by label level

dimension	level	n	cited_all	cited_Q1	cited_Q2	cited_Q3	cited_Q4
R_top	1	650	0.44	0.47	0.39	0.49	0.39
R_top	2	1820	0.55	0.55	0.48	0.57	0.58
R_top	3	619	0.60	0.63	0.57	0.56	0.62
R_top	4	851	0.61	0.61	0.58	0.60	0.65
R_top	5	900	0.63	0.64	0.59	0.63	0.65
R_ideo	2	60	0.50	0.59	0.38	0.69	0.32
R_ideo	3	1424	0.55	0.58	0.50	0.58	0.54
R_ideo	4	2806	0.56	0.55	0.53	0.57	0.60
R_ideo	5	550	0.61	0.65	0.51	0.59	0.71

3. Pull: does the RAG prediction move toward the evidence?

pull = mean party position of the chunks − baseline prediction; shift = RAG − baseline prediction. ρ > 0 and slope > 0 mean the prediction follows the evidence; slope 1 = it moves all the way.

subset	n_texts	spearman_shift_vs_pull	p	slope	share_moved_toward_evidence
all texts	829	0.59	<0.001	0.67	0.53
Q1 (RAG win)	202	0.88	<0.001	1.03	0.89
Q2 (RAG distraction)	200	0.63	<0.001	0.96	0.71
Q3 (joint failure)	216	0.42	<0.001	0.41	0.38
Q4 (baseline sufficient)	211	-0.01	0.902	0.00	0.15
cited chunks mostly relevant (R_top ≥ 4)	244	0.53	<0.001	0.61	0.50
cited chunks less relevant (R_top < 4)	585	0.62	<0.001	0.70	0.54

subset	n_texts	spearman_shift_vs_pull	p	slope	share_moved_toward_evidence
cited chunks ideological (R_ideo ≥ 4)	445	0.59	<0.001	0.67	0.52
cited chunks less ideological (R_ideo < 4)	384	0.60	<0.001	0.68	0.54

Same with all retrieved chunks (cited or not):

subset	n_texts	spearman_shift_vs_pull	p	slope	share_moved_toward_evidence
all texts	968	0.58	<0.001	0.70	0.53

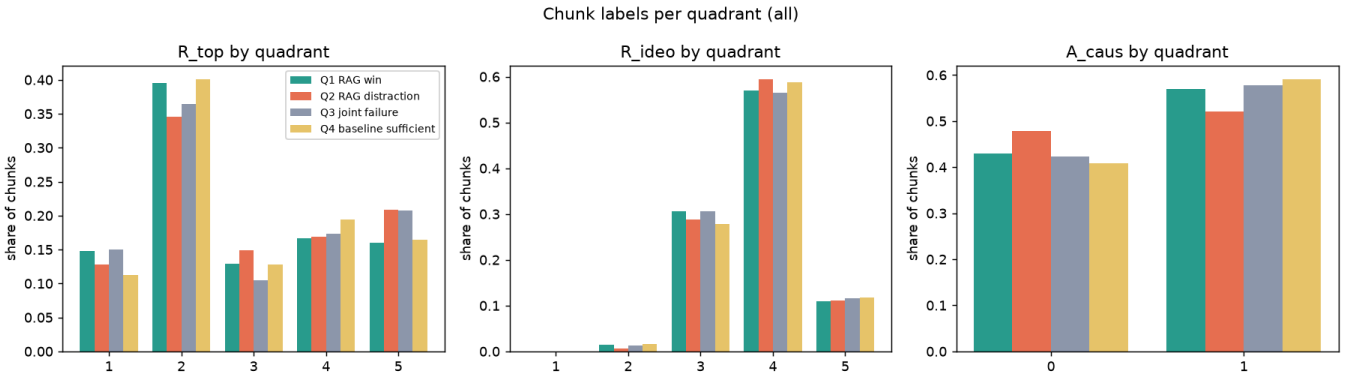
4. Evidence vs. outcome (Spearman with error reduction = base error – RAG error)

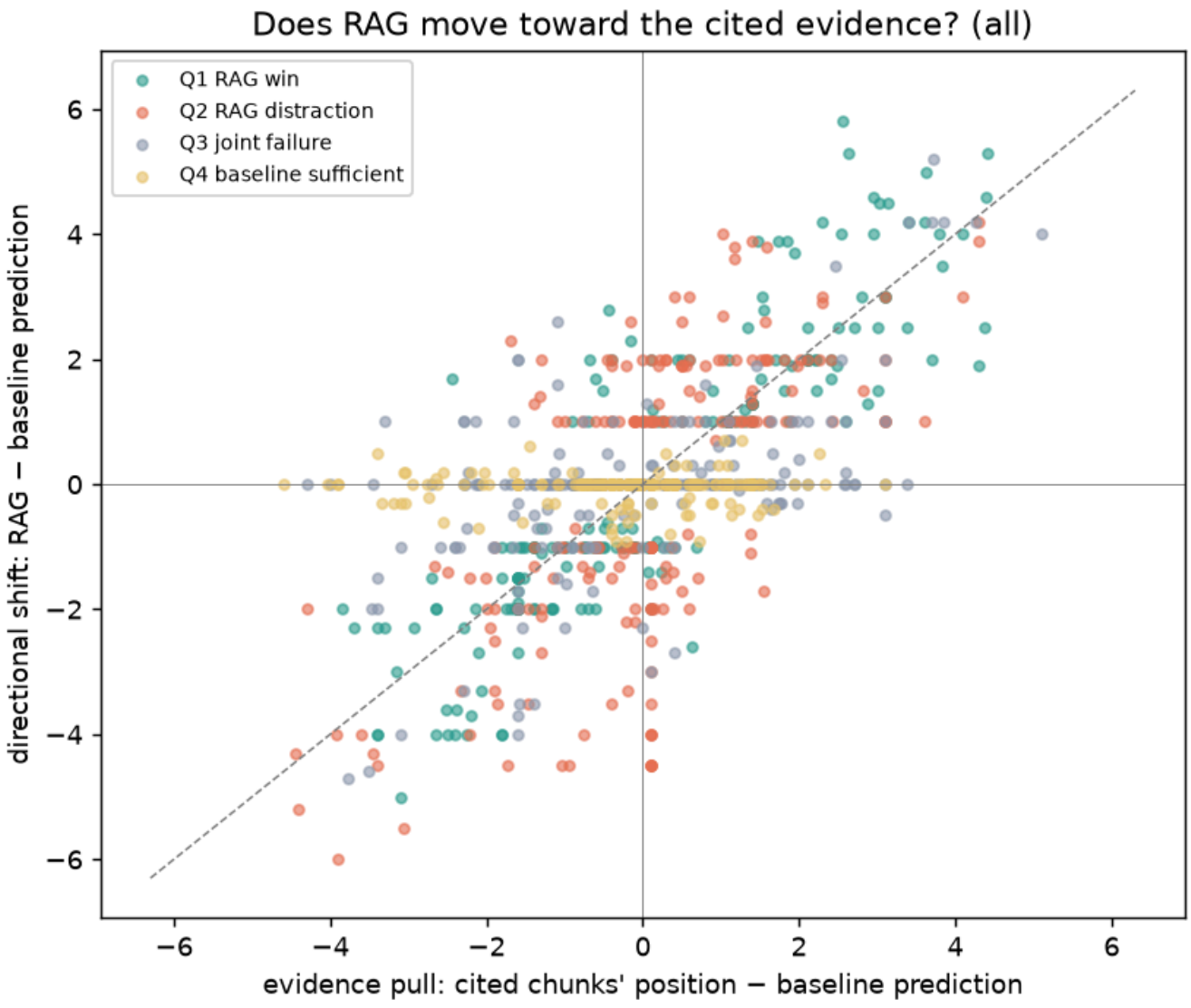
feature	n_texts	spearman_vs_error_reduction	p
cited chunks' distance to ground truth	829	-0.08	0.019
share cited chunks from input's party	831	0.11	0.002
R_top of cited chunks	831	-0.07	0.058
R_ideo of cited chunks	831	-0.04	0.245
mean R_top (all chunks)	968	-0.07	0.030
mean R_ideo (all chunks)	968	-0.05	0.125
chunks cited	968	0.04	0.268
share chunks from input's party	968	0.08	0.014

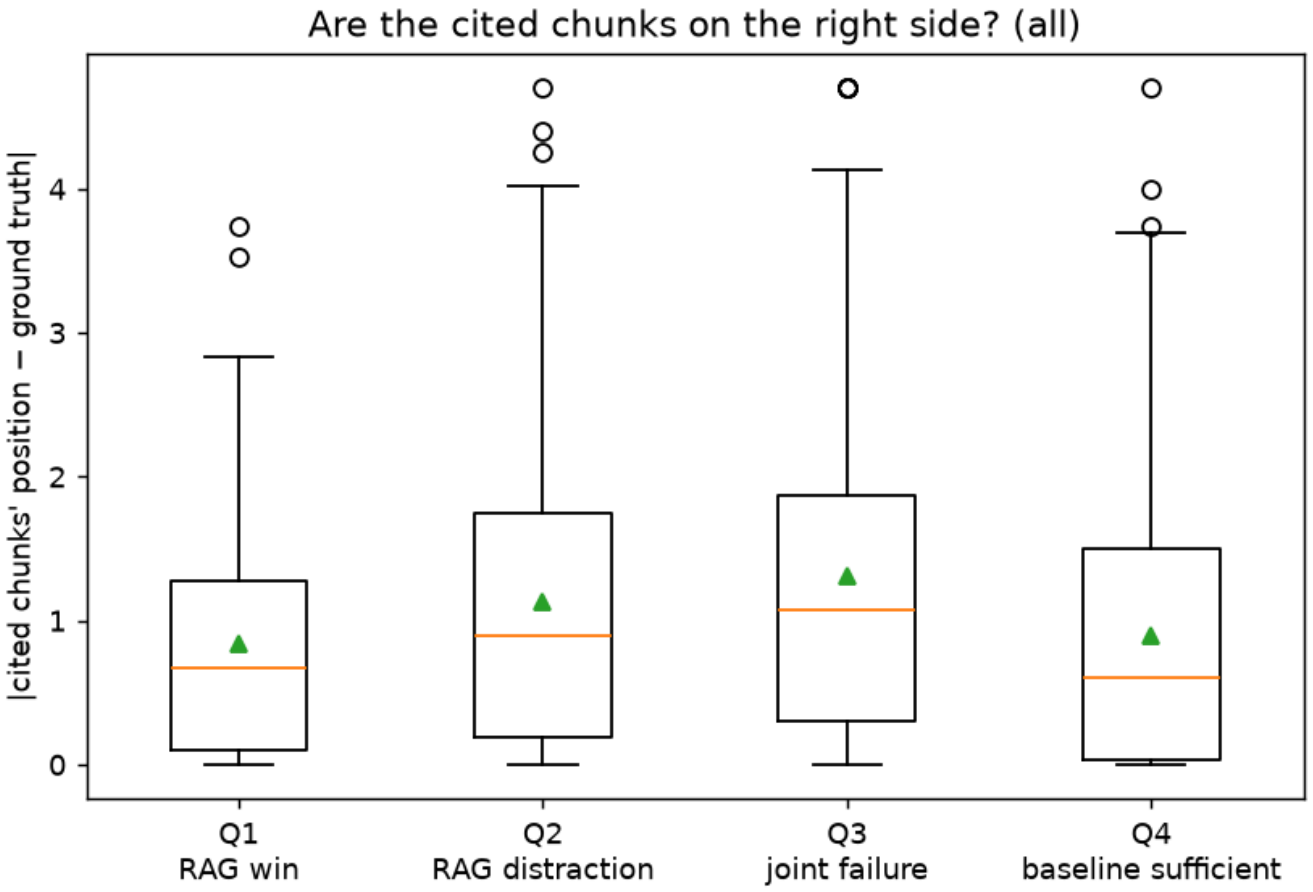
Per model (Q1 vs. Q2)

model	tier	texts	Q1 cited R_top	Q1 cited−GT	Q2 cited R_top	Q2 cited−GT	ρ(shift, pull)
llama_3b	judge	100	2.45	1.58	3.93	1.02	0.82
llama_70b	reviewed	97	2.88	0.56	3.19	0.63	0.62
llama_8b	judge	100	3.05	0.81	2.98	2.14	0.73
mistral_14b	reviewed	100	2.67	1.28	3.15	1.09	0.57
mistral_3b	judge	100	2.65	0.92	2.98	1.32	0.76

model	tier	texts	Q1 cited R_top	Q1 cited−GT	Q2 cited R_top	Q2 cited−GT	ρ(shift, pull)
mistral_8b	judge	100	2.87	1.21	3.17	1.03	0.29
qwen_32b	judge	100	3.17	0.26	3.09	1.08	0.61
qwen_3b	reviewed	71	3.10	1.26	2.56	1.25	0.49
qwen_72b	judge	100	3.03	0.37	3.09	0.67	0.62
qwen_7b	judge	100	3.41	0.70	3.14	1.35	0.34






Are the cited chunks on the right side? (all)
|cited chunks' position − ground truth|
0
1
2
3
4
Q1
RAG win
Q2
RAG distraction
Q3
joint failure
Q4
baseline sufficient